4.3 Writing Equations in Standard Form

Name _____ Period ____ Date _____

Algebra I 4.3 · Student sheet

I can…

- write equations in standard form.
- write equations of horizontal and vertical lines.
- use linear equations in standard form to solve real-life problems.

Vocabulary · fill in the missing word

Standard form — Ax + By = C, usually with integer _____ and A ≥ 0

Horizontal line — y = _____

Vertical line — x = _____

Part 1 · Ticket sales

A play earns $600 from adult tickets ($8, x) and student tickets ($5, y).

Equation: _____ x-intercept: _____ (means: _____)

y-intercept: _____ (means: _____)

Two more combinations that earn exactly $600: _____

Part 2 · Notes and examples (fill in each note, then work the example)

Standard form: Standard form: _____, with integer coefficients.

Fractions: Clear fractions by _____ every term.

Example 1

Write an equation in standard form of the line through (3, −1) with slope −2.

Example 2

Write an equation in standard form of the line through (−1, 2) and (3, 4).

Horizontal and vertical lines: Horizontal line through (a, b): _____. Vertical: _____.

Example 3

Write equations of the horizontal and vertical lines through (−4, 6).

Example 4

You spend $36 on paint ($6 per quart) and brushes ($4 each). Write an equation in standard form, and find how many brushes you buy with 4 quarts.

Part 3 · Partner practice

Partner 1

Write $y = -\frac{3}{4}x + 2$ in standard form.

Partner 2

Find A so that the graph of $Ax + 3y = 12$ passes through (2, 2).